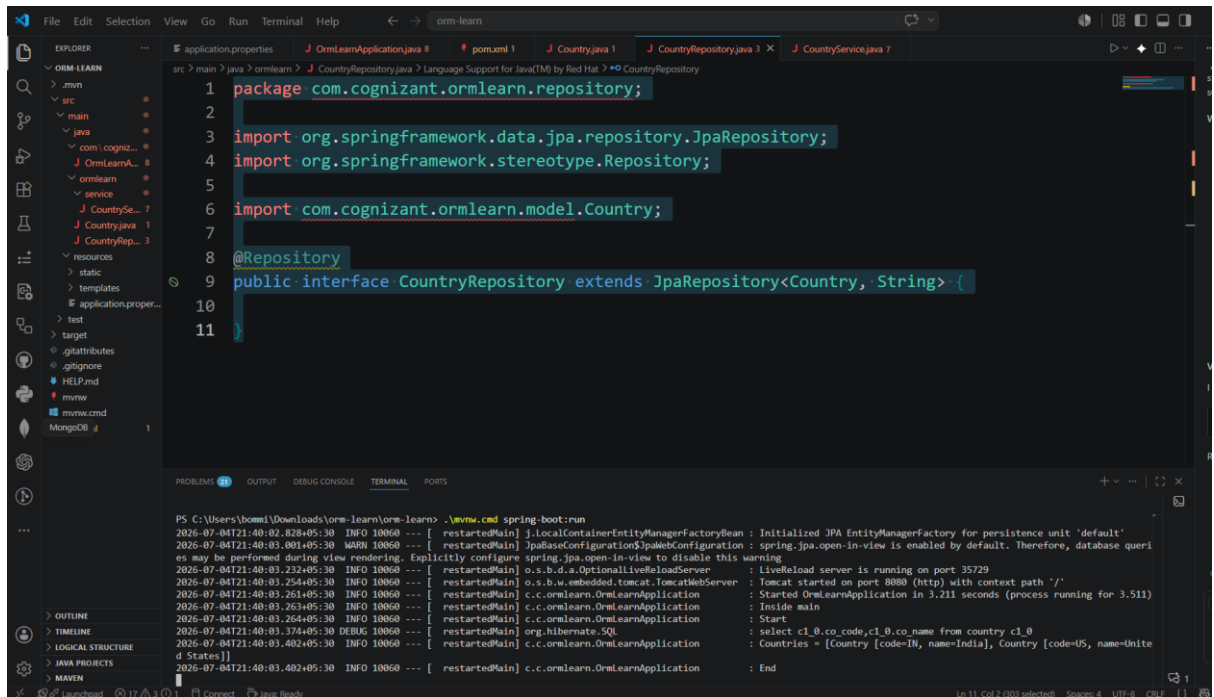


Spring Data JPA - Quick



```
1 package com.cognizant.ormlearn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5
6 import com.cognizant.ormlearn.model.Country;
7
8 @Repository
9 public interface CountryRepository extends JpaRepository<Country, String> {
10
11 }
```

PS C:\Users\booml\Downloads\orm-learn\orm-learn> .\mvnw.cmd spring-boot:run

2026-07-04T21:40:02.828+05:30 INFO 10060 --- [restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'

2026-07-04T21:40:03.001+05:30 WARN 10060 --- [restartedMain] JpaBaseConfiguration\$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning

2026-07-04T21:40:03.232+05:30 INFO 10060 --- [restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729

2026-07-04T21:40:03.254+05:30 INFO 10060 --- [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'

2026-07-04T21:40:03.261+05:30 INFO 10060 --- [restartedMain] c.c.ormlearn.OrmLearnApplication : Started OrmLearnApplication in 3.211 seconds (process running for 3.511)

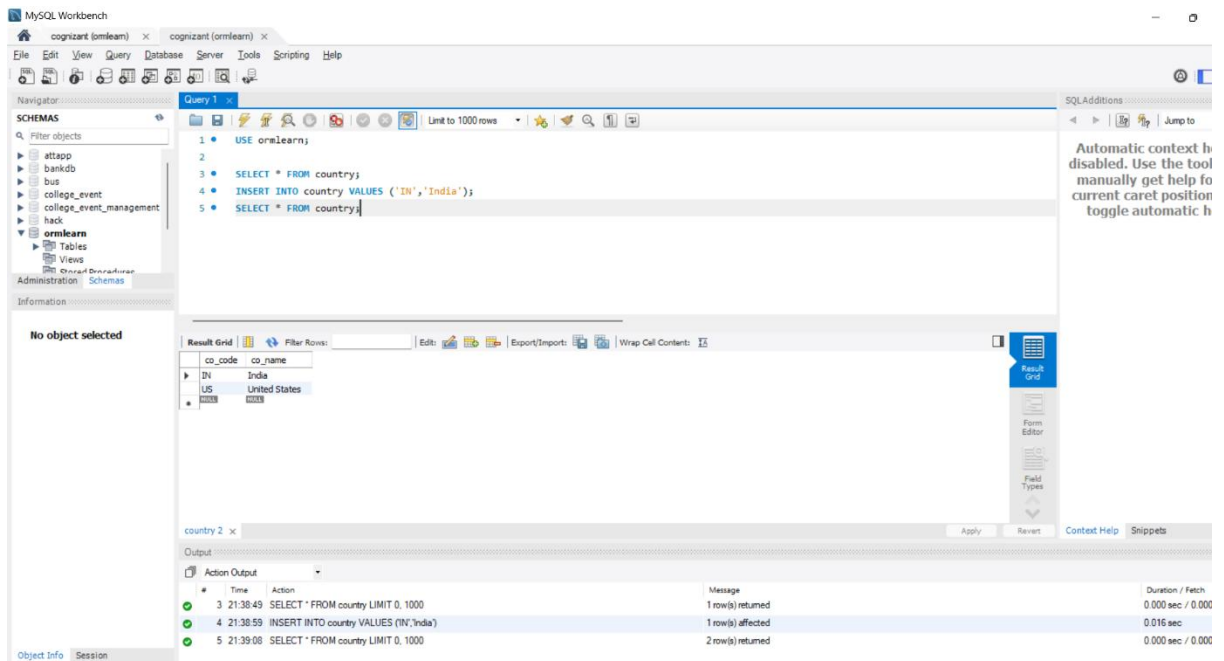
2026-07-04T21:40:03.263+05:30 INFO 10060 --- [restartedMain] c.c.ormlearn.OrmLearnApplication : Inside main

2026-07-04T21:40:03.264+05:30 INFO 10060 --- [restartedMain] c.c.ormlearn.OrmLearnApplication : Start

2026-07-04T21:40:03.374+05:30 DEBUG 10060 --- [restartedMain] org.hibernate.SQL : select c1_0.co_code,c1_0.co_name from country c1_0

2026-07-04T21:40:03.402+05:30 INFO 10060 --- [restartedMain] c.c.ormlearn.OrmLearnApplication : Countries = [Country [code=IN, name=India], Country [code=US, name=United States]]

2026-07-04T21:40:03.402+05:30 INFO 10060 --- [restartedMain] c.c.ormlearn.OrmLearnApplication : End



Automatic context help disabled. Use the tool manually get help for current caret position toggle automatic h

co_code	co_name
IN	India
US	United States

#	Time	Action	Message	Duration / Fetch
3	21:38:49	SELECT * FROM country LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000
4	21:38:59	INSERT INTO country VALUES ('IN','India')	1 row(s) affected	0.016 sec
5	21:39:08	SELECT * FROM country LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000

Difference between JPA, Hibernate and Spring Data JPA

1. JPA (Java Persistence API)

- JPA is a specification.
- A specification means a set of rules or guidelines.
- It defines how Java objects should be stored and retrieved from a database.
- Important
- JPA does not contain code to connect to the database.
- It only defines the standard way to perform database operations.

2. Hibernate

- Hibernate is an ORM (Object Relational Mapping) framework.
- It implements JPA.
- Hibernate converts Java objects into SQL statements and communicates with the database.
- Real-life example
- Using the same house example:
- Blueprint = JPA
- Workers = Hibernate
 - The workers build the house by following the blueprint.
 - Similarly, Hibernate performs the database operations by following JPA rules.

3. Spring Data JPA

- Spring Data JPA is a Spring Framework module.
- It uses JPA (and an implementation such as Hibernate) and removes most of the repetitive code.